

2019 级《线性代数 II》期终试卷(A)评分标准

下列各题均为解答题, 解答应写出文字说明、证明过程或演算步骤

$$1. \text{解: } \left\{ \begin{aligned} \frac{1}{4} |A^T A^*| &= \frac{1}{4^3} |A^T| |A^*| = \frac{1}{4^3} \cdot (-2) \cdot (-2)^{3-1} \\ &= \frac{1}{4} |A| |A^T A^{-1}| = \left(-\frac{1}{2}\right)^3 |A^T| |A^{-1}| \end{aligned} \right\} = -\frac{1}{8}.$$

$$2. \text{解: } A^2 - 3A - 4E = (A + 2E)(A - 5E) + 6E = O \Rightarrow (A + 2E) \left[-\frac{1}{6}(A - 5E) \right] = E \\ \Rightarrow (A + 2E)^{-1} = -\frac{1}{6}(A - 5E).$$

$$3. \text{解: } (\alpha_1 + k\alpha_2, \alpha_1 + 2\alpha_2 + \alpha_3, k\alpha_1 - \alpha_3) = (\alpha_1, \alpha_2, \alpha_3) \begin{pmatrix} 1 & 1 & k \\ k & 2 & 0 \\ 0 & 1 & -1 \end{pmatrix}$$

$$\text{由题设得 } \begin{vmatrix} 1 & 1 & k \\ k & 2 & 0 \\ 0 & 1 & -1 \end{vmatrix} = (k+2)(k-1) = 0 \Rightarrow k = -2 \text{ 或 } 1.$$

4. 解: 由 $n - R(A) = 1 \Rightarrow Ax = 0$ 的基础解系含一个线性无关的解向量;

$\xi = 3(\eta_1 + \eta_2) - 2(2\eta_2 + \eta_3) = (1, 0, -1, -2)^T$ 是 $Ax = 0$ 的基础解系,

$\eta^* = \frac{1}{2}(\eta_1 + \eta_2) = \frac{1}{2}(1, 2, 3, 4)^T$ 是 $Ax = b$ 的解向量,

$\Rightarrow Ax = b$ 的通解 $x = k\xi + \eta^* = k(1, 0, -1, -2)^T + \frac{1}{2}(1, 2, 3, 4)^T, k$ 为任意常数.

5. 解: A 与 B 相似 $\Rightarrow B$ 与 A 有相同的特征值 $1, \frac{1}{2}, \frac{1}{3} \Rightarrow B^{-1} + E$ 有特征值 $2, 3, 4$

$$\Rightarrow |B^{-1} + E| = 2 \cdot 3 \cdot 4 = 24.$$

$$6. \text{解: 二次型 } f \text{ 的矩阵 } A = \begin{pmatrix} t & 3 & 0 \\ 3 & 4 & 1 \\ 0 & 1 & t \end{pmatrix}$$

$$A \text{ 正定} \Rightarrow t > 0, \begin{vmatrix} t & 3 \\ 3 & 4 \end{vmatrix} = 4t - 9 > 0, |A| = t(4t - 10) > 0 \Rightarrow t > \frac{5}{2}.$$

$$7. \text{解: } (\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5) = \begin{pmatrix} 1 & 1 & 1 & 0 & 1 \\ 3 & 2 & 1 & 1 & -3 \\ 0 & 1 & 2 & 2 & 0 \\ 5 & 4 & 3 & 4 & -7 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 0 & 1 \\ 0 & -1 & -2 & 1 & -6 \\ 0 & 0 & 0 & 3 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 & 0 & -1 & 0 & -3 \\ 0 & 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (6\text{分})$$

$\Rightarrow R(\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5) = 3 \Rightarrow \alpha_1, \alpha_2, \alpha_4$ 为向量组的一个最大无关组 (2分)

$$\Rightarrow \begin{cases} \alpha_3 = -\alpha_1 + 2\alpha_2 \\ \alpha_5 = -3\alpha_1 + 4\alpha_2 - 2\alpha_4 \end{cases} \quad (2\text{分})$$

8. 解: 方程组的系数行列式为

$$|A| = \begin{vmatrix} 1 & 1 & 2-\lambda \\ 2-\lambda & 2-\lambda & 1 \\ 3-2\lambda & 2-\lambda & 1 \end{vmatrix} = (\lambda-1)^2(\lambda-3) \quad (4\text{分})$$

当 $\lambda \neq 1$ 且 $\lambda \neq 3$ 时, 方程组有唯一解; (2分)

$$\text{当 } \lambda = 3 \text{ 时, 增广矩阵 } B = \begin{pmatrix} 1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ -3 & -1 & 1 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -1 & 1 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

$\Rightarrow R(A) = 2, R(B) = 3$, 方程组无解; (2分)

$$\text{当 } \lambda = 1 \text{ 时, 增广矩阵 } B = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow R(A) = R(B) = 1 < 3$, 方程组有无穷多解, (2分)

$$\text{通解为 } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = k_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \text{ 其中 } k_1, k_2 \text{ 为任意常数. (2分)}$$

$$\text{或增广矩阵 } B \rightarrow \begin{pmatrix} 1 & 1 & 2-\lambda & 1 \\ 0 & \lambda-1 & -(\lambda-1)(2\lambda-5) & 3(\lambda-1) \\ 0 & 0 & -(\lambda-1)(\lambda-3) & \lambda-1 \end{pmatrix}, \quad (4\text{分}) \text{ 以下略.}$$

$$9. \text{解: } |B - \lambda E| = \begin{vmatrix} 4-\lambda & -6 \\ 1 & -1-\lambda \end{vmatrix} = (\lambda-1)(\lambda-2) = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = 2;$$

当 $\lambda_1 = 1$ 时, $(B - E)x = 0$ 得 $\alpha_1 = (2, 1)^T$; 当 $\lambda_2 = 2$ 时, $(B - 2E)x = 0$ 得 $\alpha_2 = (3, 1)^T$.

$$\text{记 } P = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix} \Rightarrow P^{-1}BP = A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \Rightarrow B = P\Lambda P^{-1}$$

$$\Rightarrow B^n = P\Lambda^n P^{-1} = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} -1 & 3 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 3 \cdot 2^n - 2 & 6 - 3 \cdot 2^{n+1} \\ 2^n - 1 & 3 - 2^{n+1} \end{pmatrix}. \quad (6\text{分})$$

$$\text{因 } \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^n = O, n \geq 2 \Rightarrow C^n = \left[2E + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right]^n = 2^n E + n2^{n-1} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 2^n & n2^n \\ 0 & 2^n \end{pmatrix}; \quad (3\text{分})$$

$$\Rightarrow A^n = \begin{pmatrix} B^n & O \\ O & C^n \end{pmatrix} = \begin{pmatrix} 3 \cdot 2^n - 2 & 6 - 3 \cdot 2^{n+1} & 0 & 0 \\ 2^n - 1 & 3 - 2^{n+1} & 0 & 0 \\ 0 & 0 & 2^n & n2^n \\ 0 & 0 & 0 & 2^n \end{pmatrix}. (1分)$$

10. 解: $A^2(B-E) = A+B \Rightarrow (A+E)(A-E)B = (A+E)A$

由 $|A+E| = 33 \neq 0 \Rightarrow A+E$ 可逆 $\Rightarrow (A-E)B = A$ (4分)

$$(A-E, A) = \begin{pmatrix} 2 & 0 & 1 & 3 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 & 0 \\ 1 & 0 & 1 & 1 & 0 & 2 \end{pmatrix} \xrightarrow{r} \begin{pmatrix} 1 & 0 & 0 & 2 & 0 & -1 \\ 0 & 1 & 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 & 3 \end{pmatrix} (4分)$$

$$\Rightarrow A-E \text{ 可逆}, B = (A-E)^{-1}A = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 3 \end{pmatrix}. (2分)$$

11. 解:

$$|A - \lambda E| = \begin{vmatrix} 2-\lambda & -1 & -1 \\ -1 & 2-\lambda & -1 \\ -1 & -1 & 2-\lambda \end{vmatrix} = -\lambda(\lambda-3)^2 = 0 \Rightarrow \text{特征值为 } \lambda_1 = 0, \lambda_2 = \lambda_3 = 3. (4分)$$

当 $\lambda_1 = 0$ 时, 解方程组 $Ax = 0$ 得基础解系 $\xi_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}; (2分)$

当 $\lambda_2 = \lambda_3 = 3$ 时, 解方程组 $(A-3E)x = 0$ 得基础解系 $\xi_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \xi_3 = \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} (4分)$

对正交向量组 ξ_1, ξ_2, ξ_3 单位化得正交矩阵

$$P = \begin{pmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{pmatrix}, \text{ 使 } P^{-1}AP = A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}. (2分)$$

12. 证(1) 由 $A^2 = kA, Ax = \lambda x, x \neq 0 \Rightarrow \lambda^2 x = k\lambda x \Rightarrow \lambda(\lambda - k)x = 0 \Rightarrow \lambda(\lambda - k) = 0 \Rightarrow \lambda = 0$ 或 $k; (3分)$

(2) 由 $A^2 = kA \Rightarrow A(A - kE) = O \Rightarrow R(A) + R(A - kE) \leq n;$

由 $A + (kE - A) = kE \Rightarrow R(A) + R(kE - A) \geq R(kE) = n,$

因 $R(kE - A) = R(A - kE) \Rightarrow R(A) + R(A - kE) = n; (3分)$

(3) 由于 $A = O, A = kE$ 是对角阵, 不妨设 $A \neq O$ 和 kE , 记 $R(A) = r (0 < r < n)$

$\Rightarrow Ax = 0$ 的基础解系含 $n-r$ 个线性无关的解向量, 即属于特征值 $\lambda_1 = 0, A$ 有 $n-r$ 个线性无关的特征向量;

由 $R(A - kE) = n-r \Rightarrow (A - kE)x = 0$ 的基础解系含 r 个线性无关的解向量, 即属于特征值 $\lambda_2 = k, A$ 有 r 个线性无关的特征向量.

$\Rightarrow A$ 有 n 个线性无关的特征向量 $\Rightarrow A$ 能相似对角化. (4分)